

Student Name: Class:.....



YEAR 11 PRELIMINARY ASSESSMENT

TASK 1

March 2012

**TOPICS TESTED:- Basic Arithmetic, Algebra,
Surds, Indices, Equations**

General Instructions

- Reading Time – 5 minutes
- Working Time – 60 Minutes
- Start each question on a new page.

Q 1	
Q 2	
Q 3	
Q 4	
Q 5	
TOTAL	

Question 1 (8 marks)

- (a) Express $1.2\dot{6}$ as a fraction in simplest form. 2
- (b) Evaluate $\sqrt{\frac{\pi \times 5.432^2}{18.927}}$ correct to 3 significant figures. 2
- (c) Evaluate $\frac{-4.3 \times 10^{-4} + 1.7 \times 10^{-2}}{(6.7 \times 10^6)^3}$. Express your answer in scientific notation to 3 significant figures 2
- (d) Simplify $12 - 2|5 - 8| \times 3$ 2

Question 2 (11 marks)

- (a) Factorise fully $m^2 - n^2 + 5m - 5n$ 2
- (b) Solve $\frac{3x-2}{4} - \frac{2x+1}{8} = 5$ 2
- (c) Simplify $\frac{x+1}{x^2-x} - \frac{x-1}{x^2+x}$ 3
- (d) Factorise completely $2x^3 - 128$ 2
- (e) Factorise fully $a^2 + 12ac - 28c^2$ 2

Question 3 (8 marks)

- (a) Simplify $\sqrt{75} + \sqrt{48} - \sqrt{12}$ 2
- (b) Express with a rational denominator $\frac{\sqrt{5}+1}{2\sqrt{2}+3}$ 2
- (c) Find integers a and b such that $(3 - \sqrt{2})^2 = a - b\sqrt{2}$ 2
- (d) Form a pair of simultaneous equations and solve them to find x and y
 $6 + \sqrt{x-y} = x + y + \sqrt{18}$ 2

Question 4 (10 marks)

- (a) Find the exact value of $\frac{A^2C}{B^2}$ where $A = \left(\frac{2}{3}\right)^2$ $B = \left(\frac{4}{3}\right)^4$ $C = \left(\frac{8}{3}\right)^2$ 3
- (b) Simplify $\frac{16}{2^{3x} \times 8^{1-x}}$ 2
- (c) Simplify $\frac{x^{-1} + y^{-1}}{x + y}$ 2
- (d) Simplify $\frac{x - 5 + 6x^{-1}}{1 - 2x^{-1}}$ 3

Question 5 (17 marks)

- (a) Solve $|2x - 3| \leq 5$. 2
- (b) Solve $|x + 2| = 2x - 5$. 3
- (c) Find the exact values of x if $2x^2 - 4x + 1 = 0$ 2
- (d) Solve the following simultaneous equations
- i) $x + y = 5$ and $x^2 + y^2 = 25$ 2
- ii) $2x + y = 4$ and $5x + 2y = 9$ 2
- iii) $y = x^2 - 3x + 4$ and $y = 2x - 2$ 2
- (e) Solve $-12 \leq \frac{2x}{3} < 4$ and graph on a number line 2
- (f) Solve $(x - 3)^2 = 25$ 2

END OF ASSESSMENT

Solutions

Question 1

$$a) 10x = 2.666...$$

$$x = 0.266...$$

$$9x = 2.4$$

$$x = \frac{2.4}{9}$$

$$= \frac{24}{90}$$

$$\therefore 1.26 = 1 \frac{24}{90} = 1 \frac{8}{30}$$

$$b) \sqrt{4.8976485}$$

$$= 2.21 \text{ (3 sf)}$$

$$c) 5.51 \times 10^{-23}$$

$$d) 12 - 2 \times 3 \times 3$$

$$12 - 18$$

$$-6$$

Question 2.

$$a) (m-n)(m+n) + 5(m-n)$$

$$= (m-n)(m+n+5)$$

$$b) 2(3x-2) - (2x+1) = 40$$

$$6x-4-2x-1=40$$

$$4x = 45 \rightarrow$$

$$c) \frac{x+1}{x(x-1)} - \frac{x-1}{x(x+1)}$$

$$= \frac{(x+1)^2 - (x-1)^2}{x(x-1)(x+1)}$$

$$= \frac{x^2+2x+1-x^2+2x-1}{x(x-1)(x+1)}$$

$$= \frac{4x}{x(x-1)(x+1)}$$

$$= \frac{4}{(x-1)(x+1)}$$

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$$d) 2(x^3-64)$$

$$2(x^3-4^3)$$

$$2(x-4)(x^2+4x+16)$$

$$e) (a+14c)(a-2c)$$

QUESTION 3.

$$a) \sqrt{75} + \sqrt{48} - \sqrt{12}$$

$$5\sqrt{3} + 4\sqrt{3} - 2\sqrt{3}$$

$$7\sqrt{3}$$

$$b) \frac{\sqrt{5}+1}{2\sqrt{2}+3} \times \frac{2\sqrt{2}-3}{2\sqrt{2}-3}$$

$$\frac{2\sqrt{10}+2\sqrt{2}-3\sqrt{5}-3}{8-9}$$

$$= \frac{2\sqrt{10}+2\sqrt{2}-3\sqrt{5}-3}{-1}$$

$$= -2\sqrt{10}-2\sqrt{2}+3\sqrt{5}+3$$

$$c) (3-\sqrt{2})^2 = 9+2-6\sqrt{2}$$

$$= 11-6\sqrt{2}$$

$$a=11 \quad b=6$$

$$d) x+y=6$$

$$x-y=18$$

$$2x=24$$

$$x=12$$

$$y=-6$$

QUESTION 4.

$$a) \left(\frac{2}{3}\right)^4 \times \left(\frac{8}{3}\right)^2 \div \left(\frac{4}{3}\right)^8$$

$$= \left(\frac{2^4}{3^4} \times \frac{2^6}{3^2} \times \frac{3^8}{2^{16}}\right)$$

$$= \frac{3^2}{2^6} = \frac{9}{64}$$

$$b) \frac{2^4}{2^{3x} \times 2^{3-3x}} = \frac{2^4}{2^3} = 2.$$

$$c) \frac{x^{-1} + y^{-1}}{x + y}$$

$$= \frac{\frac{1}{x} + \frac{1}{y}}{x + y}$$

$$= \frac{y + x}{xy} \times \frac{1}{x + y}$$

$$= \frac{1}{xy}$$

$$d) \frac{x - 5 + \frac{6}{x}}{1 - \frac{2}{x}} = \frac{x^2 - 5x + 6}{x - 2}$$

$$= \frac{(x-3)(x-2)}{(x-2)}$$

$$= x - 3$$

QUESTION 5.

$$a) -5 \leq 2x - 3 \leq 5$$

$$-2 \leq 2x \leq 8$$

$$-1 \leq x \leq 4.$$

$$b) x + 2 = 2x - 5 \quad -x - 2 = 2x - 5$$

$$7 = x \quad \text{or} \quad -3x = -3$$

$$x = 7 \quad \text{or} \quad x = 1$$

check solutions

$$|x + 2| = 2x - 5 \quad |1 + 2| = 2 \times 1 - 5$$

$$|9| = 14 - 5 \quad 3 = -3$$

$$= 9 \quad \text{False}$$

true

only $x = 7$ a solution

$$c) x = \frac{4 \pm \sqrt{16 - 8}}{4}$$

$$= \frac{4 \pm \sqrt{8}}{4}$$

$$= \frac{4 \pm 2\sqrt{2}}{4} = \frac{2 \pm \sqrt{2}}{2}$$

$$d) y = 5 - x \quad x = 0 \quad x = 5$$

$$x^2 + (5 - x)^2 = 25 \quad y = 5 \quad y = 0$$

$$x^2 + 25 + x^2 - 10x = 25$$

$$2x^2 - 10x = 0$$

$$ii) \begin{array}{r} 4x + 2y = 8 \\ 5x + 2y = 9 \\ \hline -x = -1 \\ \therefore x = 1 \end{array}$$

sub in (1)

$$2 + y = 4$$

$$y = 2$$

$\therefore x = 1 \quad y = 2$

$$iii) 2x - 2 = x^2 - 3x + 4$$

$$0 = x^2 - 5x + 6$$

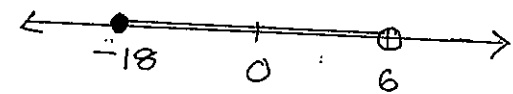
$$(x - 3)(x - 2) = 0$$

$$x = 3 \quad x = 2$$

$$y = 4 \quad y = 2$$

$$e) -36 \leq 2x < 12$$

$$-18 \leq x < 6$$



$$f) (x - 3) = \pm 5$$

$$x = 3 \pm 5$$

$$x = 8 \text{ or } -2$$